

JKBOSE 2026

CLASS 11TH

RESPIRATION IN PLANTS

LECTURE-01

BOTANY

ARSHID BHAT SIR



Topics To Covered

- 1** Introduction
- 2** Respiratory Substrate
- 3** Types Of Respiration

Introduction



Metabolism

It is Sum total of all the Chemical reactions which occur inside a living cell / organism.

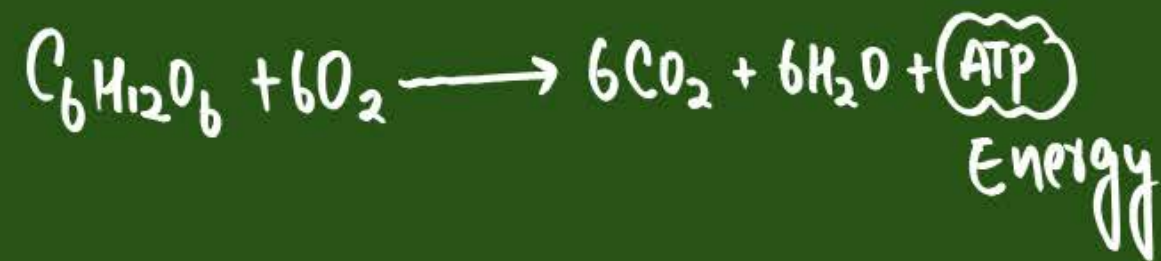
Anabolism

- Build-up rxns.
- Endothermic rxns
or
Endergonic rxns.
- Example: photosynthesis



Catabolism

- Break down rxns.
- Exothermic rxns.
or
Exergonic rxns.
- Example: Respiration



* Respiration is :

- a. Catabolic process.
- b. Energy releasing process
- c. Step wise oxidative breakdown of food
- d. Multistep process.
- e. Enzymatically controlled process.
- f. temperature dependent process.

≡ The basic and ultimate aim of respiration is to produce energy from food.



≡ Each and Every requires energy to perform different activities.



≡ Thus food gets transferred to each and every cell where its respiration takes place.



≡ Thus actual respiration is called as **cellular respiration**.

≡ During respiration, Energy is released from breaking of C-C bond present in food (ex. $C_6H_{12}O_6$)

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≡ Energy releases in a Stepwise manner / Controlled manner (not in a single step)

↓

≡ Energy released during respiration gets stored in the form ATP.
(Adenosine triphosphate).

↓

≡ During Energy demand in the cell, ATP comes into action. Thus ATP is called as **Energy currency of cell.**

Respiratory Substrate



- ≡ Respiratory Substrates are organic molecules which take part in respiration.
- ≡ Common respiratory Substrates are :
 - Carbohydrates
 - Proteins
 - Fats
 - Organic acid (Malic acid, Oxaloacetic acid).
- ≡ Most common respiratory Substrate = Carbohydrates.
- ≡ Among Carbohydrates, most common respiratory Substrate = Simple Sugars like Glucose (Hexose Sugar)
 $C_6H_{12}O_6$
- ≡ Carbohydrates are instant sources of Energy.

NOTE

- Water, Vitamins, Minerals — are not source of Energy.

Types Of Respiration

A. On the basis of respiratory substrate.

B. On the basis of presence / Absence of Oxygen.

A. ON the Basis of Respiratory Substrate, respiration is of two types:

①. Protoplasmic respiration:

- ≡ Here respiratory substrate used are proteins.
- ≡ It occurs under starvation conditions.
- ≡ Cell does not continue this respiration for long because the end products of proteins is toxic ammonia (may cause cell death).
- ≡ Occurs during germination of seeds, spores...

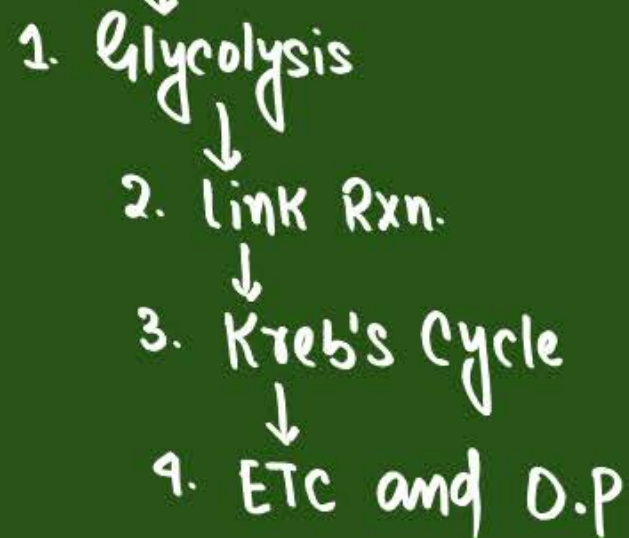
②. Floating respiration:

- ≡ Here respiratory substrate used are carbohydrates / fats.

B. On the basis of presence / Absence of O_2 , respiration is of 02 types:

① Aerobic respiration

- ≡ In presence of O_2 .
- ≡ Complete Oxidation (breakdown) of food occurs.
- ≡ Example: Eukaryotic organisms.
- ≡ Mechanism: $C_6H_{12}O_6 \rightarrow$ Respiratory Substrate.

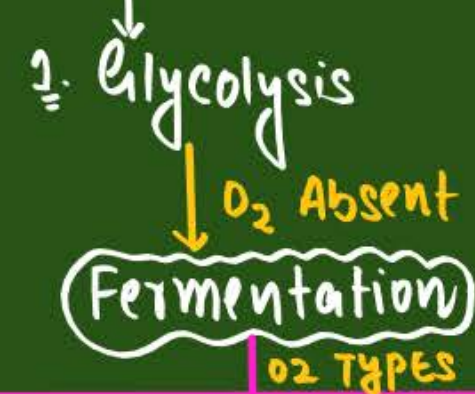


≡ End products: $6CO_2$, H_2O , 38 ATP...

② Anaerobic respiration

- In absence of O_2 .
- Incomplete / partial Oxidation (breakdown) of food occurs.
- Example: Mostly Prokaryotes.

≡ Mechanism: $C_6H_{12}O_6 \rightarrow$ Respiratory Substrate



≡ End products: CO_2 , Ethanol (C_2H_5OH), lactic acid, 02 ATP.

QUESTIONS

How is the energy present in food released during biological oxidation?

- A** In one step, quickly
- B** In two steps, slowly
- C** In different steps, slowly
- D** In different steps, quickly

QUESTIONS

ATP is formed in

- A** Respiration
- B** Photosynthesis
- C** Photophosphorylation
- D** All the above

QUESTIONS

What is wrong about respiration?

- A** It does not occur in cell
- B** Oxidation occurs without the use of enzymes
- C** Energy is released in one step quickly
- D** All the above ni ai zieyloyld

HOMework

Revision

💡 Do plants Breathe?

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Do it yourself...

Thank
You